

Extensional PERs*

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A class of Partial Equivalence Relations (PERs) is described such that the resulting *full* subcategory of the realizability universe has the expected properties of a good category of CPOs; that is, it is a Cartesian closed category and every endomorphism has a canonical fixed point. Moreover, the reflection functor into the subcategory of strict maps (usually called the "lifting operation") yields a good notion of "partial map," a necessary condition if one wishes to maintain a connection between strict maps and partial maps. It is shown that all functors that arise in practice have canonical invariant objects; hence a host of domain equations are guaranteed to have solutions. © 1992 Academic Press, Inc.

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The search for good categories of domains in an effective setting has occupied the attention of many. Early on it was recognized that the existence of strong categorical closure properties in conjunction with effectiveness provided a framework for producing powerful computational models describing the mathematical semantics of programming languages (Scott, 1981). This idea was reinforced with the introduction of the effective (or realizability) and recursive topoi which utilized tools from topos theory to create new categorical models of computation (Hyland, 1982; Mulry, 1982). These topoi created mathematical universes where the notions of realizability and recursiveness existed intrinsically as part of their internal structure. A further important development was the observation that the notion of domain itself could be defined internally inside (Rosolini, 1986). Thus not only the domains themselves, but also any maps between such domains, would be effective. Later it became apparent that the subcategory of modest sets or, as here treated, PERs inherited enough properties from the realizability topos, including internal completeness, to make it an alternative candidate for interpreting domains (Carboni, Freyd, and Scedrov, 1987).

Since PERs are partial equivalence relations on the natural numbers, the category *PER* is more accessible and more easily describable using familiar tools from computation theory such as Turing machines. *PER*, however, is not without fault; even defining the notion of partial order on a PER requires further refinement. The obvious step is then to restrict further to PERs with additional structure. In this paper the search is for a class of PERs such that the resulting full subcategory has the expected properties of any good category of CPOs: it should be a CCC (Cartesian closed category) and every endomorphism should have a canonical fixed point. Moreover, the reflection functor (usually called the "lifting operation") should yield a good notion of "partial map." The last condition was the most elusive of all. It is necessary if one wishes to maintain a connection between strict maps and partial maps. A fine point is that these various functors should live in the realizability universe: it is that point that ensures the existence of a host of domain equation solutions.

1. CONVENTIONS

Let us fix notation. If A is a PER (*Partial Equivalence Relation on the natural numbers*) then $x\{A\}y$ means that x and y are related by A , $x:A$ means $x\{A\}x$, and $[x:A]$ is the equivalence class of x as defined by A .

There are two fundamental binary operations on PERs each of which requires a binary operation on natural numbers. For PERs A and B define $A \times B$ by $\langle x, y \rangle \{A \times B\} \langle u, v \rangle$ iff $x\{A\}u$ and $y\{B\}v$, where $\langle , \rangle$ denotes

a pairing function on the natural numbers. Define $A \Rightarrow B$ by $m\{A \Rightarrow B\}n$ iff $mx\{B\}ny$ whenever $x\{A\}y$, where mx denotes the result of a binary partial operation to be interpreted as the number written on the tape when the m th Turing machine comes to a halt after having been started with x as initial tape. We use the convention that whenever a catenation appears in a formula it is to be understood that the catenation in question is defined. Thus $mx\{B\}ny$ means that the m th turing machine with initial tape x comes to a halt, that the n th turing machine starting with initial tape y comes to a halt, and that the final tapes of the two machines are equivalent according to the PER B .

Further conventions: we write $m.x\{B\}n.y$ to mean that either both mx and ny are undefined or $mx\{B\}my$. (We think of $m.x$ as something of an inner product, undefinedness being orthogonality). In the absence of parentheses we turn on the machines in the order written: xyz means $(xy)z$. We use 0 to denote both the empty tape and the empty PER. We understand that $0x$ is never defined, that $1x = x$, $kxy = x$, $ex.y = y.x$ all x, y (k and e are understood to be specific natural numbers). We use $x;y$ to name the machine such that $(x;y).z = y.(x.z)$ all z . $(x;)$ denotes the machine such that $(x;)y = x;y$. We also use 1 to denote the grossest PER: $x\{1\}y$ all x, y (hence $1:A \Rightarrow 1$ all A). N , for "natural numbers," denotes the identity relation (i.e., $x\{N\}y$ iff $x = y$). The category **PER** has PERs as objects. The maps from A to B are of the form $[m:A \Rightarrow B]$. The composition of $[m:A \Rightarrow B]$ and $[n:B \Rightarrow C]$ is $[(m;n):A \Rightarrow C]$. The identity map of A is $[1:A \Rightarrow A]$. **PER** is, of course, a CCC and it has all finite limits and colimits.

2. PARTIAL-MAP CLASSIFIERS

The standard categorical definition of partial map yields far too many partial maps in **PER** (there are far too many subobjects). There is, however, a ready notion of "natural domain" for a partial function and it arises from the natural notion of partial map. We say that m describes a partial map from A to B if $mx\{B\}my$ whenever $x\{A\}y$ and mx is defined. By our conventions this says that $x\{A\}y$ and mx defined implies my defined. The domain of this partial map is a union of A -classes. Not any old union, but one that arises by intersecting the domain of A with a recursively enumerable subset of the natural numbers. These subobjects of A will be called herein, **NATURAL SUBOBJECTS** and it will be understood that all things to be called **PARTIAL MAPS** have natural subobjects as domains.

PROPOSITION. *For every PER B there is a PER, ΣB , such that the partial*

maps from A to B are in natural correspondence with the maps from A to ΣB .

(ΣB is said to “classify” the partial maps into B .) ΣB is the PER defined by: $m\{\Sigma B\}n$ iff $m.0\{B\}n.0$. Then $[k:B \Rightarrow \Sigma B]$ is an isomorphism onto a natural subobject of ΣB , to wit, all $m:\Sigma B$ such that $m0$ is defined. The partial map from ΣB back to B that stands as the (partial) inverse of k is named by $e0$. This partial map is the **UNIVERSAL, PARTIAL MAP**. That is, given any partial map (with, we repeat, a natural domain) from A to B there is a unique total map from A to ΣB that when followed by the universal partial map is the given partial map.

Composition of partial maps is constructed via pullbacks along their domains; hence this translates to the following (in which partial maps are unmentioned):

If A' is a natural subobject of A and $A' \rightarrow B$ is a map there exists a unique map $A \rightarrow \Sigma B$ such that

$$\begin{array}{ccc} A' & \longrightarrow & A \\ \downarrow & & \downarrow \\ B & \longrightarrow & \Sigma B \end{array}$$

is a pullback.

Σ which is short for $\Sigma 1$ is the equivalence relation (not just partial equivalence relation) that partitions the natural numbers into just two classes: $[0:\Sigma]$ and $[1:\Sigma]$. m is in $[1:\Sigma]$ iff the m th Turing machine halts when started with the empty tape. As a special case of the last proposition we have:

COROLLARY. *Maps into Σ classify natural subobjects; that is, if $[1:A' \Rightarrow A]$ is a natural subobject of A then there exists a unique map $A \rightarrow \Sigma$ such that*

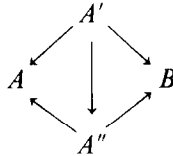
$$\begin{array}{ccc} A' & \longrightarrow & A \\ \downarrow & & \downarrow \\ 1 & \longrightarrow & \Sigma \end{array}$$

is a pullback.

The lower map is the **UNIVERSAL NATURAL SUBOBJECT** $[k:1 \Rightarrow \Sigma]$.

(It may be worthwhile to remember the standard definition of a partial

map from A to B : it is named by a monomorphism $A' \rightarrow A$ (the name of its "domain") and an arbitrary map $A' \rightarrow B$; another such pair $\langle A'' \rightarrow A, A'' \rightarrow B \rangle$ names the same partial map if there is an isomorphism $A' \rightarrow A''$ such that



For every function (in the category of sets) from N to N there is a partial map (in the category of PERs) with a dense domain (that is, a subobject that meets every non-trivial subobject). If we identify two partial maps with dense domains if their equalizer is still dense the resulting monoid is naturally equivalent with the ordinary monoid of all functions from N to N .)

3. ExPER's

We define m and n to be **ABSOLUTELY COEXTENSIONAL** if $m.s = n.s$ all s . We need a more relative notion. For S a subset of the natural numbers define m and n to be **COEXTENSIONAL MOD S** if $m.s = n.s$ all s in S . A PER, A , is **EXTENSIONAL MOD S** if for all $m:A$ and all n it is the case that $m\{A\}n$ iff m and n are coextensional mod S . (Note that this says, in particular, that if m and n are coextensional mod S then $m:A$ iff $n:A$.) We will say that a PER, A , is an **EXTENSIONAL PER** (**ExPER** for short) if there is some subset, S , such that A is extensional mod S . In fact, one does not have to search for S : for any PER, A , take its **EXTENSIONAL BASE**, $Base(A)$, to be $\{s \mid a.s = b.s \text{ whenever } a\{A\}b\}$. A is an ExPER iff it is extensional mod $Base(A)$. (If A is non-empty and extensional mod S then, in fact, S must be $Base(A)$.)

The full subcategory of ExPERs is cartesian closed but not without a little work. The construction for products we gave for general PERs does not work: it does not yield ExPER's. First define a binary operation $[,]$ on Turing machines by taking $[a, b]$ to be a machine such that $[a, b].s =$ if $[s \text{ even}]$ then $a.(s \text{ DIV } 2)$ else $b.(s \text{ DIV } 2)$. Suppose that A and B are ExPER's. Define $A \times B$ to be that ExPER with extensional base $2(Base(A)) \cup (2(Base(B) + 1)$ just large enough so that $[a, b]:A \times B$ whenever $a:A$ and $b:B$. (The construction we gave for the empty product, 1, as the maximal equivalence relation yields an ExPER. Its extensional base is the empty set.)

We have the same sort of problem with exponentiation as with products: if B is an ExPER we define $A \Rightarrow B$ to be the ExPER with extensional base $\{\langle a, s \rangle \mid a:A \text{ and } s \text{ in } \text{Base}(B)\}$ so that $m:A \Rightarrow B$ iff $[m.\langle a, s \rangle = m.\langle a', s \rangle \text{ whenever } a\{A\}a' \text{ and } s \text{ in } \text{Base}(B)]$ and [for any $a:A$ there exists $b:B$ such that for all s in $\text{Base}(B)$ it is the case that $m.\langle a, s \rangle = b.s$].

It is not difficult to give explicit isomorphisms between these constructions and the previous constructions. Such not only establishes the Cartesian closedness of the full sub-category of ExPERs but shows that the inclusion functor preserves the structure. (Note that we did not need A to be an ExPER for the construction of $A \Rightarrow B$: the full subcategory of ExPER's is an "exponential ideal" in the realizability universe.) The construction of equalizers is straightforward and arbitrary products from the point of view of the realizability universe are also easy (the intersection of a collection of ExPER's is extensional mod the union of the corresponding collection of subsets.)

THEOREM. *The full subcategory of ExPER's is reflective in the category of PERs.*

To construct the reflection of an arbitrary PER, A , define $A!$ to be the ExPER with extensional base $\{m \mid m.a = m.a' \text{ whenever } a\{A\}a'\}$ just large enough so that $ea:A!$ for all $a:A$. The reflector map is $[e:A \Rightarrow A!]$. $A!$ is a functor: let $f:A \Rightarrow B$. Note first that $(f;)$ yields a function from $\text{Base}(B!)$ to $\text{Base}(A!)$. Define $f!$ by taking $(f!)n$ to be a machine such that $(f!)n.m = n.(f;m)$. We must show that if A is an ExPER then $[e:A \Rightarrow A!]$ is an equivalence, indeed, with $[(e;):A! \Rightarrow A]$ as its inverse. First verify that $(e;)(ea)$ and a are absolutely coextensive for any a . Second show that $(e;):A! \Rightarrow A$. It follows immediately that $(e; (e;))\{A \Rightarrow A\}1$. Using the fact that $[e:A \Rightarrow A!]$ is an epimorphism (by construction of $A!$) we obtain $((e;);e)\{A! \Rightarrow A!\}1$. The fact that $[e:A \Rightarrow A!]$ is a reflection of A into the full subcategory of ExPER's is now a formality: given $f:A \Rightarrow B$ where B is an ExPER we obtain a factorization of f through e from the diagram

$$\begin{array}{ccc}
 A & \xrightarrow{e} & A! \\
 f \downarrow & & \downarrow f! \\
 B & \xrightarrow{e} & B! \\
 & \searrow 1 & \downarrow (e;) \\
 & & B
 \end{array}$$

The uniqueness of the factorization follows from the fact that $[e:A \Rightarrow A!]$ is an epimorphism.

4. ExPERs AS DOMAINS

Any ExPER has an canonical partial ordering: if A is an ExPER and $m, n:A$ then we define $m \leq n:A$ if m is the result of restricting the convergence of n to a subset of $Base(A)$, that is, $m \leq n:A$ iff $ms = ns$ whenever s an element of $Base(A)$ such that ms is defined. A remarkable fact is

THEOREM. *Any map between ExPER's is order-preserving.*

We show that the existence of a counterexample allows us to solve the halting problem. The proof is assembled from three lemmas. First we show that for any ExPER A and $m < n:A$ there is a map $a:\Sigma \Rightarrow A$ such that $a0\{A\}m$ and $a1\{A\}n$. Next we show that for $p, q:B$ such that it is *not* the case that $q \leq p:B$, there is a map $b:B \Rightarrow \Sigma$ with $bp\{\Sigma\}0$ and $bq\{\Sigma\}1$. Given a counterexample $f:A \Rightarrow B$, that is, given $m < n:A$ such that it is not the case that $fm \leq fn:B$, we can use these two lemmas to construct an endomorphism on Σ that transposes 0, 1. The third lemma says that such an endomorphism allows us to solve the halting problem.

For the first lemma suppose that $m < n:A$ and define $a:\Sigma \Rightarrow A$ by taking ax to be a machine that upon hearing input s attempts to compute $x0, ms$ and ns in parallel; if ms converges then ax halts with ms as output; if both $x0$ and ns converge then ax halts with ns as output.

For the second lemma suppose that $p, q:B$ are such that it is not the case that $q \leq p:B$. Choose s in $Base(B)$ so that qs is defined but it is not the case that $qs = ps$ (possibly because ps is undefined). Define $b:B \Rightarrow \Sigma$ by $br.0 = \text{if}[rs = qs] \text{ then } 0$ (else undefined). (br is a machine whose only output is 0; it halts upon hearing 0 as input only if rs halts and $rs = qs$.)

For the third lemma suppose $t:\Sigma \Rightarrow \Sigma$ were such that $t0\{\Sigma\}1$ and $t1\{\Sigma\}0$. Then t would allow us to solve the halting problem: given any machine m run $m0$ and $tm0$ until one of them halts; which one halts decides the issue. ■

Given a set S of natural numbers we let $AS(S \Rightarrow \Sigma N)$ denote the object of **ASCENDING SEQUENCES** in $S \Rightarrow \Sigma N$, defined by $c:AS(S \Rightarrow \Sigma N)$ iff for all n it is the case that $cn:S \Rightarrow \Sigma N$ and $cn \leq c(n+1):S \Rightarrow \Sigma N$. Given $c:AS(S \Rightarrow \Sigma N)$ we understand $sup(c)$ to be a machine such that $sup(c)s$ is defined for s in S iff cns is defined for some n and in that case $sup(c)s = cns$. (We can define $sup(c)$ to be a machine that upon hearing input s attempts to compute in parallel $c0s, c1s, c2s, \dots$; then $sup(c)s$ halts iff cns halts for some n ; if $sup(c)s$ halts then the output is determined by the "first" cns to halt. The notion of "first," of course, is up to the designer of the machine. Note, however, that the notion of first does not affect the result for ascending sequences.)

If B is an ExPER we let $AS(B)$ be the restriction of the PER $AS(Base(B) \Rightarrow \Sigma N)$ to those $c:AS(Base(B) \Rightarrow \Sigma N)$ that take values in B , that is, those c for which $cn:B$ all n . We say that B is **COMPLETE** if for every every $c:AS(B)$ it is the case that $\sup(c):B$. Note that every ExPER of the form $A \Rightarrow \Sigma N$ (for A any PER) is complete.

THEOREM. *Any map between ExPER's is continuous.*

We wish to show that if $f:A \Rightarrow B$ and A and B are ExPER's then $f(\sup(a))$ is the supremum of $a;f$ whenever $a:AS(A)$ and $\sup(a):A$. We will show that the existence of a counterexample allows us to construct a decision algorithm to determine when a recursive function $N \rightarrow N$ has a zero. Suppose $a:AS(A)$ and $f:A \Rightarrow B$ yield a counterexample; that is, suppose $\sup(a;f) < f(\sup(a))$. If we enlarge B to be the maximum ExPER with the same extensional base we still have a counterexample. We assume that B is so enlarged; in particular we assume that $\sup(a;f):B$. Using the second lemma from the last proof we may find a map $b:B \Rightarrow \Sigma$ such that $b(\sup(a;f))\{\Sigma\}0$ and $b(f(\sup(a)))\{\Sigma\}1$. The composition $f;b$ still yields a counterexample.

To recapitulate, we may suppose that $a:AS(A)$ and $g:A \Rightarrow \Sigma$ are such that $g(an)0$ is undefined for all n but $g(\sup(a))0=0$. Given $h:N \Rightarrow N$ we construct an element $c:A$ with the property that $gc0$ is defined iff h has no zeros (which clearly allows us to decide whether h has a zero). We take c to the machine such that cs is defined iff for some m , $[h0, h1, \dots, h(m-1)]$ are all positive] and $[ams$ is defined] and in that case $cs=ams$. Then $c\{A\} \sup(a)$ if h has no zeros; otherwise $c\{A\} an$ for n the first zero of h . (To be sure that c is constructible from h and a consider a machine that upon hearing input s attempts to find n such that ans is defined; if there is no such n then cs will be undefined; if there is such n then the machine searches for the smallest $m < n$ such that $hm=0$; if there is no such m then $cs=ans$; if there is such m then $c.s=am.s$.) ■

PROPOSITION. *The full subcategory of complete ExPER's is reflective in the category of PERs.*

The proof is similar to the proof that the category of all ExPER's is reflective. The biggest change is in the definition of $A!$: keep the definition of $Base(A!)$ as $\{m \mid m.a = m.a' \text{ whenever } a\{A\}a'\}$ but define $A!$ to be the intersection of all *complete* ExPER's with that extensional base and which allow the map $e:A \Rightarrow A!$. No change is needed in the definition of $f!$. To verify that e is still an epimorphism note that the equalizer of any pair of maps from a complete ExPER to any ExPER is itself complete. ■

5. REFLECTIVITY OF STRICT MAPS

Let $\mathbf{Exp}$ denote the full subcategory of complete $\mathbf{ExPER}$'s, and $\mathbf{Exp}_0$ the further full subcategory of $\mathbf{ExPER}$'s that include 0 as an element. Let $\mathbf{Exp}_1$ denote the lluf subcategory of $\mathbf{Exp}_0$ of strict maps. We state the theorems in this section and the next for the case of complete $\mathbf{ExPER}$'s because that is where we will use them, but it may be noted that the proofs do not require completeness.

THEOREM. $\Sigma: \mathbf{Exp} \rightarrow \mathbf{Exp}_1$ is a reflector functor.

Just as for products and exponentiation, we must reconstruct the Σ -functor for $\mathbf{ExPER}$'s: If A is an $\mathbf{ExPER}$ let ΣA be the $\mathbf{ExPER}$ whose extensional base is the union of the one-element set $\{0\}$ and the set $1 + \text{Base}(A)$; define $m: \Sigma A$ iff either $[m$ is everywhere undefined on $\text{Base}(\Sigma A)]$ or $[m0 = 0$ and $(\text{succ}; m): A]$. (succ is the successor map: $(\text{succ})n = n + 1$.) It is routine that this ΣA is isomorphic to the previous construction and that it is complete if A is. (Note that $\text{Base}(1)$ is the empty set and that under the $\mathbf{ExPER}$ construction $\text{Base}(\Sigma)$ is $\{0\}$. $m: \Sigma$ iff $m0$ is either undefined or equal to 0.) Define $q: A \Rightarrow \Sigma A$ by $qm.s = \text{if } [s = 0] \text{ then } 0 \text{ else } m.(s - 1)$. The behavior of Σ on maps is defined by $(\Sigma f)a.s = \text{if } [s = 0] \text{ then } a.0 \text{ else } fa.(s - 1)$.

If $0: A$ we obtain

$$\begin{array}{ccc} A & \longrightarrow & \Sigma A \\ & \searrow & \downarrow (\text{succ};) \\ & & A. \end{array}$$

1

which together with the functor-action gives the existence property for a reflection. The uniqueness property is easy. ■

Let $\mathbf{Exp}_2$ be the further lluf subcategory of **TOTALLY STRICT** maps, those that not only preserve but *reflect* bottom (that is, those with the property that the inverse image of bottom is bottom). Σ maps $\mathbf{Exp}$ into $\mathbf{Exp}_1$, but lands, in fact, in $\mathbf{Exp}_2$. It is not an equivalence of categories: it does not preserve the terminator, indeed:

PATHOLOGY. $\mathbf{Exp}_2$ does not have a terminator.

$\mathbf{Exp}_2$ does have two *subterminators*, to wit, 1 and Σ . That is, as can be easily shown, every $\mathbf{ExPER}$ with bottom allows at most one totally strict map to either 1 or Σ . (Indeed, 1 is a *strict coterminator* in $\mathbf{Exp}_2$.) If A is an $\mathbf{ExPER}$ with bottom and with exactly one equivalence class of elements bigger than bottom we can use the first paragraph of the proof of the first

theorem of Section 4 to obtain totally strict maps $a:\Sigma \Rightarrow A$ and $b:A \Rightarrow \Sigma$ which are easily shown to form an isomorphism pair. If A has more than one equivalence class of elements bigger than bottom then there is more than one totally strict map of the form $a:\Sigma \Rightarrow A$, hence A can not be a subterminator.

But Σ is not a terminator; that is, not every object of $\mathbf{Exp}_1$ has a totally strict map to Σ . Consider for any subset S of natural numbers the maximal $\mathbf{ExpER}$, $\hat{S}$, with S as base. (Be prepared to use, therefore, that $x:\hat{S}$ all x .) It may be shown that there is a totally strict map from $\hat{S}$ to Σ iff S is recursively enumerable. The direction we need here is the following: suppose $b:\hat{S} \Rightarrow \Sigma$ is totally strict; let c be defined by $cn.s = \text{if } [s=n] \text{ then } 0 \text{ (else undefined)}$; then S is the domain of definition of the recursive partial map named by $c;b;(e0)$. ■

We may, as always, factor the functor $\mathbf{Exp} \rightarrow \mathbf{Exp}_2$ through the appropriate slice category to obtain $\mathbf{Exp} \rightarrow (\mathbf{Exp}_2/\Sigma) \rightarrow \mathbf{Exp}_2$, where the second functor is the forgetful functor. Since Σ is a subterminator in $\mathbf{Exp}_2$, $\mathbf{Exp}_2/\Sigma \rightarrow \mathbf{Exp}_2$ is a full embedding. The objects of $\mathbf{Exp}_2/\Sigma$ are pairs $\langle A, n \rangle$ where A is a complete $\mathbf{ExpER}$ such that $0:A$ and $n:A \Rightarrow \Sigma$ with $[n:A \Rightarrow \Sigma]$ being totally strict. The full embedding $\mathbf{Exp} \rightarrow \mathbf{Exp}_2/\Sigma$ sends B to $\langle \Sigma B, 1 \rangle$.

THEOREM. *$\mathbf{Exp} \rightarrow \mathbf{Exp}_2/\Sigma$ is an equivalence of categories.*

The required functor $\mathbf{Exp}_2/\Sigma \rightarrow \mathbf{Exp}$ sends $\langle A, n \rangle$ to the $\mathbf{ExpER}$ with the same base as A just large enough to contain all but the bottom of A . The resulting endofunctor on $\mathbf{Exp}$ sends an $\mathbf{ExpER}$, A , to A' , where A' is the $\mathbf{ExpER}$ with base equal to $1 + \text{Base}(A)$ as large as can be so that $(suc;):A' \Rightarrow A$. It is easily verified that $(suc;)$ is an equivalence of functors. The resulting endofunctor on $\mathbf{Exp}_2/\Sigma$, on the other hand, sends $\langle A, n \rangle$ to $\langle A', 1 \rangle$ where A' is the $\mathbf{ExpER}$ with the same base as ΣA as large as can be so that $qn:A' \Rightarrow A$ where $qna.s = \text{if } [s=0] \text{ then } na.0 \text{ else } a.(s-1)$. (Technical note: $\mathbf{Exp}_2/\Sigma$ as a category object in the effective topos does not have an object of objects that is codiscrete, or as sometimes said, uniformizable. The natural transformation named by q , in particular, is not constant as we will require in Section 8 for a realizable transformation on $\mathbf{PER}$.)

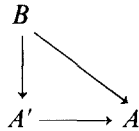
(From the point of view of the realizability universe this last theorem means that the "categorical completion" of $\mathbf{Exp}_0$ is constructible as a subcategory. That is, if we regard $\Sigma:\mathbf{Exp}_0 \rightarrow \mathbf{Exp}_2/\Sigma$ as an enlargement of $\mathbf{Exp}_0$ we may take it to be the reflection of $\mathbf{Exp}_0$ into a suitable category of complete categories.)

6. MULTI-COREFLECTIVITY OF STRICT MAPS

ExP not only should be reflective in the realizability universe (which ensures a host of completeness conditions) but as detailed by the first-named author at the 1989 New Orleans MFPS conference, the further subcategory of strict maps, ExP_1 , should be both reflective and “multi-coreflective”.

THEOREM. *ExP_1 is multi-coreflective in ExP .*

The notion of “multi-,” due to Diers, specializes to the following: the **MULTI-COREFLECTION** (in ExP_1) of an object A (in ExP) is a collection of maps of the form $A' \rightarrow A$, where each A' is in ExP_1 , with the property that for each map $B \rightarrow A$ with B in ExP_1 there is a unique triangle of the form

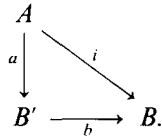


such that the horizontal arrow is in the multi-coreflection and the vertical map is in ExP_1 . It is easy to see that each $A' \rightarrow A$ in the multi-coreflection is determined by what it does to the bottom of A' , indeed that the equivalence classes of A index the multi-coreflection. It is then easy to see that for each A -class we seek A' in ExP_1 with an isomorphism to the principal updeal in A with the given class as bottom. That may be achieved as follows: Let A be an ExPER. Given $b:A$ we construct A' to be an ExPER whose extensional base is $Base(A) \times N \setminus \{\langle s, j \rangle \mid bs = j\}$; in words, the set of all ordered pairs where the first coordinate is in $Base(A)$ except for those ordered pairs that compose the graph of the partial function named by b . Stipulate $m:A'$ iff there is $n:A$ such that $b \leq n:A$ and $m.\langle s, j \rangle =$ if $[ns = j]$ then 0 (else undefined). It is easy to verify that $0:A'$. The map $q:A' \Rightarrow A$ is defined by $qm.s = \mu_j [m.\langle s, j \rangle = 0]$ or $[bs = j]$. (For this to work in the realizability universe several points must be checked. The ExPER A' depends on $[b:A]$ but not on b . The name for the map $A' \rightarrow A$ does depend on b but the map itself depends only on $[b:A']$. Given $x:B \Rightarrow A$, where $0:B$ and $x0\{A\}b$, define $fx:B \rightarrow A'$ by $fx.y\langle i, j \rangle =$ if $[xyi = j]$ then 0 (else undefined). f does not depend on A, B, x, b . ■

COROLLARY. *Idempotents split in ExP_0 .*

Given an idempotent $i:A \Rightarrow A$ in ExP_0 let $1:B \Rightarrow A$ be an equalizer of

1, $i: A \Rightarrow B$ in **ExP**. From a multi-coreflection of B let $b: B' \Rightarrow B$ be the map that allows a factorization of $i: A \Rightarrow B$:

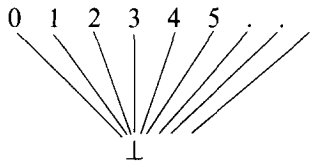


Then a and b split i . ■

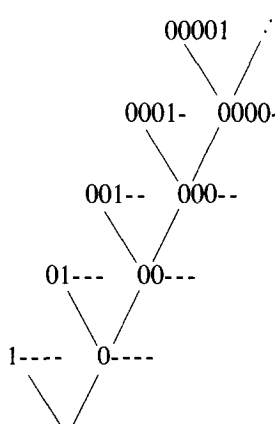
7. THE EXTENSIONAL NATURAL NUMBERS

As described in Freyd (1990), there are several varieties of natural numbers objects in a category such as **ExP**₁, each for different mapping problems.

To begin with, the construction of N we have been using is not extensional but—conveniently— ΣN is. $Base(\Sigma N)$ is the one-element set $\{0\}$ and ΣN is the largest ExPER with that base. If we remove the bottom we obtain a natural numbers object for **ExP**. With the bottom we have the **HORIZONTAL NATURAL NUMBERS** object, $FlatN$, for **ExP**₁:



In **ExP**₁ the **OBLIQUE NATURAL NUMBERS**, $LazyN$, is most conveniently defined by taking all of N as its extensional base and including all machines whose output sequences look like



where “---” means “undefined.” There is a machine at the top that delivers the everywhere-zero sequence.

The canonical map $LazyN \rightarrow FlatN$ is given by a machine that listens to the output of a $LazyN$ -machine, counts the number of zeros, halts upon hearing a one, and outputs the number of zeros.

The **VERTICAL NATURAL NUMBERS** (or “ordered natural numbers”), $OrdN$, will be taken to be the family of those machines in $LazyN$ that emit only zeros:

$$\begin{array}{c} \vdots \\ | \\ 3 \\ | \\ 2 \\ | \\ 1 \\ | \\ 0 \end{array}$$

(Do not forget the machine at the top that delivers the everywhere-zero sequence.)

The canonical map $LazyN \rightarrow OrdN$ may be described as the result of following the $LazyN$ machines with a filter, one that passes only zeros.

$LazyN$ is the initial $\Sigma \vee \Sigma X$ -algebra. A $\Sigma \vee \Sigma A$ -algebra structure on an object A may be construed as a chosen point $p:A$ and a self-map $t:A \Rightarrow A$ (neither of which has to be strict). For $LazyN$ we take $p=-----$ and t to be defined by $tx.s = \text{if } [s=0] \text{ then } 0 \text{ else } x.(s-1)$. The induced map $f:LazyN \rightarrow A$ may be defined by taking fx (for $x:LazyN$) to be a machine which upon hearing input s in $Base(A)$ searches for n such that either $[xn=1 \text{ and } t^n ps \text{ is defined}]$ or $[xn=0 \text{ and } t^{n+1}0s \text{ is defined}]$; such n being found, $fxs = \text{if } [xn=1] \text{ then } t^n ps \text{ else } t^{n+1}0s$; if there is no such n then $fx.s$ is undefined.

$OrdN$ is the initial $1 \vee \Sigma X$ -algebra. A $1 \vee \Sigma A$ -algebra structure on an object A may be construed as a special case of a $\Sigma \vee \Sigma A$ -algebra structure, to wit, one where $p\{A\}0$. The same recipe works for the induced map $OrdN \rightarrow A$, indeed, it may be viewed as the result of factoring the previously defined map through $LazyN \rightarrow OrdN$.

$FlatN$ is the initial $\Sigma \vee X$ -algebra. A $\Sigma \vee A$ -algebra structure on an object A may be construed as a special case of a $\Sigma \vee \Sigma A$ -algebra structure, to wit, one where $t:A \Rightarrow A$ is strict (but p is arbitrary). The induced map $f:FlatN \rightarrow A$ may be defined by taking fx (for $x:FlatN$) to be a machine which upon hearing input s in $Base(A)$ tries to compute $x0$; if successful its output is $t^{x0}ps$.

We have only began. As for initial algebras so it is for final coalgebras.

LazyN is the final $\Sigma \vee \Sigma X$ -coalgebra. As argued in Freyd (1990), a $\Sigma \vee \Sigma A$ -coalgebra structure on an object A may be construed as a natural subobject A' of A together with a *partial* self-map $t:A \Rightarrow \Sigma A$ subject to the condition that A' and the domain of t are disjoint. The induced map $f:A \rightarrow \text{Lazy}N$ may be defined by $fx.n = \text{if } t^n x:A' \text{ then } 1 \text{ else if } t^{n+1}x \text{ is defined then } 0 \text{ (else undefined)}$.

OrdN is the final $1 \vee \Sigma X$ -coalgebra. A $1 \vee \Sigma A$ -coalgebra structure on an object A may be construed as a special case of a $\Sigma \vee \Sigma A$ -coalgebra structure, to wit, one where A' is empty. The same recipe works for the induced map $A \rightarrow \text{Ord}N$.

FlatN is the final $\Sigma \vee X$ -coalgebra. A $\Sigma \vee A$ -coalgebra structure on an object A may be construed as a proper natural subobject A' of A together with a strict self-map $t:A \Rightarrow A$ subject to the condition that t restricted to A' is constantly 0 (i.e., $a:A'$ implies $ta\{A\}0$). For the induced map $A \rightarrow \text{Flat}N$ consider the machine which upon hearing input $x:A$ seeks n such that $t^n x:A'$. Under the stipulated conditions there is at most one such n . Such a machine delivers a partial map $A \rightarrow N$, therefore a total map $A \rightarrow \text{Flat}N$ (since $\Sigma N = \text{Flat}N$).

(In this last process we seem to be throwing away partial information being obtained on lower bounds for n . Under the present conditions, however, that partial information may be unavailable: since A' is not decidable there may, for example, be no moment when we know that x is not in A' until we have found a positive n such that $t^n x$ is in A' .)

(We need not require full decidability. Suppose that A'' is a natural subobject that mimics the complement of A' in two ways: it is small enough to be disjoint from A' and large enough to "cover the support of t ," that is, large enough so that for every $a:A$ either $ta\{A\}0$ or $a:A''$. Then we may replace t with a partial map whose domain is A'' and lift $A \rightarrow \text{Flat}N$ to a map $A \rightarrow \text{Lazy}N$. The more inclusive the A'' the better the partial information, or what is the same thing, the higher the induced map is in the intrinsic ordering of $A \Rightarrow \text{Lazy}N$.)

8. DOMAIN EQUATIONS

A functor, T , on the category **PER** is a **REALIZABLE FUNCTOR** if there is a machine t such that $t1 = 1$ and for all $x:A \Rightarrow B$ it is the case (assuming that T is covariant) that $tx:TA \Rightarrow TB$, indeed it is the case that $T[x:A \Rightarrow B] = [tx:TA \Rightarrow TB]$. To put into words: though there is no condition on what T does to objects its action on maps must be computable and the only way the action on maps can be computable is if $T[x:A \Rightarrow B]$

depends only on x (not on A or B); moreover T must preserve identity maps in the strong sense.

At first glance the condition seems to be prohibitive. It was first noted as the “externalization” of the condition that the functor exist in the realizability universe (as a functor on the category object **PER**). Having been noted, the realizability universe may be ignored if one checks that basic functors are realizable and that basic constructions preserve realizability (essentially guaranteed by the fact that the realizability universe is a topos).

A **REALIZABLE TRANSFORMATION** between realizable functors is a natural transformation such that there is a machine n such that $n:TA \Rightarrow SA$ names the transformation for all A . Again it seems prohibitive but in fact is the condition that the transformation lives in the realizability universe, hence may be expected to hold.

If T is a covariant realizable functor then T preserves the containment relation on PERs: the condition that $1:A \Rightarrow B$ is the same as the condition that A —viewed as a set of ordered pairs—is contained in B (since $t1 = 1$ we have that $1:A \Rightarrow B$ implies $1:TA \Rightarrow TB$). The objects of $\mathbf{Exp}_0$ form a complete lattice; hence there is a least fixed-object: $TA = A$. There is no a priori reason to believe such to be a minimal fixed-object; indeed, if $\mathbf{Exp}_0$ is replaced with $\mathbf{Exp}$ or the entire category, **PER**, the question is open. However,

PROPOSITION. *For covariant realizable functors on $\mathbf{Exp}_0$ least fixed-objects are minimal invariant objects.*

Before the proof we state two other propositions.

PROPOSITION. *Suppose that T is a contravariant realizable functor on $\mathbf{Exp}_0$. Let A be the least fixed-object of the covariant functor TT . Then TA is the greatest fixed-object and we know that $1:A \Rightarrow TA$. Indeed, $1:A \Rightarrow TA$ is an isomorphism and it supplies the structure to make A into a minimal T -invariant object.*

Suppose now that T is a realizable bifunctor on $\mathbf{Exp}_0$, contravariant on the first variable, covariant on the second. Define the contravariant functor F by taking FA to be the least object such that $FA = TA(FA)$ and taking $F[x:A \Rightarrow B]$ to be the least element in (FB, FA) such that $Fx = Tx(Fx)$. Define G to be at least object such that $G = F(FG)$. Then $G = T(FG)G$ and $1:T(FG)G \Rightarrow TGG$ (since $T11 = 1$). That is, G is contained in TGG .

PROPOSITION. *$1:G \Rightarrow TGG$ is an isomorphism and it supplies the structure to make G into a minimal T -invariant object.*

The proofs rely on the fact that $\mathbf{Exp}_0$ is a CPO-category as defined in Freyd (1990). Actually this statement is literally true only if interpreted from the point of view of the realizability universe; that is, the hom-sets are not CPO's in the ordinary sense but in the sense that *recursive* ascending sequences have suprema. Better to say—if one wishes to ignore the realizability universe—that all “relevant” ascending sequences have suprema; that composition preserves the suprema of relevant sequences as do the relevant functors (the realizable functors).

We will use the fact that the least fixed-object, A , of a covariant realizable functor, T , is the least “pre-fixed-object,” that is, $1:A \Rightarrow B$ whenever $1:TB \Rightarrow B$. Viewing $\langle A, 1 \rangle$ as a T -invariant object we wish to show, in the language of Freyd (1990), that it is a “special” T -invariant object. Accordingly let $e:A \Rightarrow A$ be the minimal invariant idempotent. Let B be its equalizer constructed as the extensional PER with the same extensional base as A such that $x:B$ iff $ex\{A\}x$. (The minimality of e ensures that $0:B$.) We may factor $[e:A \Rightarrow A]$ as $[e:A \Rightarrow B]$ followed by $[1:B \Rightarrow A]$ and note that composition in the other order, that is, $[1:B \Rightarrow A]$ followed by $[e:A \Rightarrow B]$, is the identity map on B . If we apply T we obtain that $1:TB \Rightarrow A$. Since $[1:TB \Rightarrow TA]$, followed by $[e:TA \Rightarrow TB]$, is the identity map on TB it is the case that $x:TB$ implies $ex\{TB\}x$, hence $ex\{A\}x$. Thus $x:TB$ implies $x:B$ and B is a pre-fixed-object and $A = B$. Hence for all $x:A$ it is the case that $ex\{A\}x$ and $e\{A \Rightarrow A\}1$. That is, the least invariant idempotent is the identity map, which by definition says that $\langle A, 1 \rangle$ is a special T -invariant object, which according to Freyd (1990) says that it is minimal invariant (and both an initial T -algebra and a final T -coalgebra).

For the contravariant case we follow the proof of the covariant-functors-suffice theorem of Section 8 in Freyd (1990). We know that the structure is supplied by the *unique* map $b:A \Rightarrow TA$ such that

$$\begin{array}{ccc} TTA & \xrightarrow{TTb} & TTTA \\ \downarrow 1 & & \downarrow 1 \\ A & \xrightarrow{b} & TA \end{array}$$

Clearly if $b = 1$ this diagram commutes, hence by uniqueness it must be the case that $b = 1$. The bifunctor case is dispatched in a similar manner.

9. INTRINSIC DESCRIPTIONS

All of the categorical structure discussed so far is intrinsic to the Cartesian closed category $\mathbf{Exp}_0$ together with its “fixed-point structure.” First define the dinatural transformation $\gamma:(X \Rightarrow X) \rightarrow X$ by $\gamma f = \sup(f^n 0)$.

Dinaturality in this case means that for every $A \rightarrow B$

$$\begin{array}{ccc}
 & (A \Rightarrow A) \longrightarrow A & \\
 (B \Rightarrow A) \swarrow & & \searrow \\
 & (B \Rightarrow B) \longrightarrow B &
 \end{array}$$

(Note that in the special case that $A = B$ we may start with the identity map in the left-most corner to obtain $f(yf) = yf$.)

The bottom-point $k0:1 \rightarrow X$ is definable as $y1_X$. We may then define the subcategory $\mathbf{Exp}_1$ as the subcategory of all objects and just those maps that preserve bottom-points. ΣX is then definable as the reflection into $\mathbf{Exp}_1$.

The canonical ordering on any object, A , is constructible as a subobject of $A \times A$, to wit, the subobject named by the monomorphism from $\Sigma A \Rightarrow A$ to $A \times A$ obtained by evaluating on the two points of Σ .

If $\mathbf{Exp}_0$ is viewed as a small category in the realizability universe then there is only one transformation (dinatural or not) from $X \Rightarrow X$ to X .

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REFERENCES

- CARBONI, A., FREYD, P. J., AND SCEDROV, A. (1987), A categorical approach to realizability and polymorphic types, in "Proceedings of the Third ACM Workshop on Mathematical Foundations of Programming Language Semantics," Lecture Notes in Computer Science, Springer-Verlag, Berlin.
- FREYD, P. (1990), Recursive types reduced to inductive types, in "Proceedings, 5th Annual IEEE Symposium in Logic and Computer Science, Philadelphia, June 1990."
- HYLAND, J. M. E. (1982), The effective topos, in "L. E. J. Brouwer, Centenary Symposium" (A. S. Troelstra and D. Van Dalen, Eds.), North-Holland, Amsterdam.
- MULRY, P. S. (1982), Generalized Banach-Mazur functionals in the topos of recursive sets, *J. Pure Appl. Algebra* **26**.
- ROSOLINI, G. (1986), "Continuity and Effectiveness in Topoi," Ph.D. thesis, University of Oxford.
- SCOTT, D. (1981), "Lectures on a Mathematical Theory of Computation," Technical Monograph PR6-19, Oxford University.